ASEAN
GEOAI FUSION 2026

TRAINING MODULE

Machine Learning Applications in Geospatial Analysis

GIS, GeoAI, and the algorithms reshaping how we read the planet



What you'll be able to do

- **Describe** machine learning in geospatial analysis.
- **Discuss** the evolution of GIS with machine learning.
- **Describe** examples of applying machine learning in GIS.
- **Discuss** the advantages of machine learning in GIS.

01

ML in geospatial

02

Evolution of GIS

03

Application examples

04

Advantages

How the module is structured

- **1.0** Introduction — what ML in geospatial analysis is.
- **2.0** Evolution of GIS — from static maps to AI-integrated smart systems.
- **3.0** Applications of ML in GIS — examples by domain.
- **4.0** Advantages of ML in GIS — accuracy, efficiency, prediction, scale.
- **5.0** Summary — key takeaways and next steps.

Module path

You are here → 1.0 Introduction

SECTION 1.0

Introduction

What ML in geospatial analysis is, and the data it works on.

ML in geospatial analysis

Definition

The application of algorithms that enable computers to learn from and make predictions based on geospatial information.

What it enables

- **Detect** hidden spatial patterns.
- **Automate** image analysis at scale.
- **Predict** future trends from spatial and temporal data.
- **Integrate** vast amounts of raster and vector data.

By integrating ML into GIS, organizations enhance their spatial analysis — enabling more informed decisions.

Typical problems and ML methods

Common geospatial analysis problems and how ML addresses them:

Spatial prediction / interpolation	Deterministic, geostatistics, machine learning
Modelling with uncertainty	Geostatistics, machine learning (measurement errors)
Multivariate joint prediction	Co-kriging, multi-task learning
Risk mapping	Indicator kriging, simulations, Mixture Density Networks
Optimisation of monitoring networks	Space-filling models, kriging, Support Vector Machines
Data mining in high-dimensional feature space	Supervised and unsupervised learning algorithms

ML excels in high-dimensional geo-feature spaces where classical methods struggle.

SECTION 2.0

The Evolution of GIS

From static maps to smart, AI-driven systems.

Static maps → smart systems

PAST

Static Maps

GIS was used to create static maps and analyse spatial data — primarily visualization and basic analytics.

TODAY

Smart Cities

AI-powered GIS platforms monitor traffic patterns, manage utilities, and optimise emergency responses in real time.

FUTURE

AI-integrated GIS

Faster data processing, predictive modelling, and real-time decision-making — redefining how we map the world.

INDUSTRY PERSPECTIVE

"AI and ML are redefining mapping technologies — faster data processing, predictive modelling, and real-time decision-making."

— Jack Dangermond, Esri UC 2024 plenary

SECTION 3.0

Applications of ML in GIS

Four domains where ML is already reshaping outcomes.

Four core domains



Urban Planning

Population density, land use, resource allocation, and development decisions.



Environmental Monitoring

Land-use change, deforestation tracking, and climate impact assessment via image segmentation.



Transportation & Logistics

Route optimisation and traffic pattern analysis for faster, leaner movement.



Disaster Management

Predicting and managing natural disasters by combining historical and real-time data.

Predictive modeling & forecasting

Identify patterns and trends — predict outcomes.

In agriculture:

- Analyse soil quality, weather, and crop health.
- Predict yields before the season ends.
- Anticipate drought conditions early.

Beyond agriculture:

- Urban demand forecasting
- Infrastructure load prediction
- Climate and weather scenarios

EXAMPLE WORKFLOW

1. **Collect data**
2. **Train model**
3. **Forecast outcome**
4. **Decide action**

Image recognition & object detection

Process and analyse satellite imagery automatically.

- **Detect changes** in land use, vegetation, urban growth.
- **Monitor deforestation** patterns over time.
- **Protect endangered species** by mapping habitats.
- **Identify illegal activity** — e.g. logging, mining, fishing.

IN THE FIELD

Conservationists use GeoAI systems to monitor deforestation patterns and protect endangered species habitats.

These systems automatically identified illegal logging activity in near real time — replacing slow, manual image review with continuous, scalable analysis.

Automation & real-time monitoring

AUTOMATION & PROCESS OPTIMIZATION

- **Automate** data cleaning, spatial analysis, mapping updates.
- **Disaster management** — evacuation maps in minutes, not hours.
- **Faster response** when every minute counts.
- **Repeatable workflows** that scale across regions and teams.

REAL-TIME MONITORING & DECISION SUPPORT

- **Integrate IoT sensors** with GIS and AI for live data.
- **Traffic management** — process live traffic and adjust signals.
- **“Metro City” case** — significantly reduced congestion.
- **Decisions** are made with the freshest data, not yesterday's snapshots.

SECTION 4.0

Advantages of ML in GIS

Accuracy, efficiency, prediction, and scale — together.

Why ML + GIS is a powerful combination

Enhanced Accuracy

Minimise human error by leveraging large datasets — insights that manual methods miss.

Increased Efficiency

Automate processing of billions of features across vast areas — faster insights, time saved.

Predictive Capabilities

Anticipate future trends from historical data — proactive decisions in planning and disaster response.

Scalability

Handle massive datasets easily — essential for environmental science and urban development.

KEY TAKEAWAYS

ML turns GIS from a mapping tool into a predictive, decision-making platform — at the speed and scale the modern world requires.